

Que Let integral of $f(x) = \frac{x^{2009}}{(1+x^2)^{1006}}$ w.r.t x is equal to $\frac{1}{n} \left(\frac{x^2}{1+x^2} \right)^m + C$, then find $m+n$.

Soln

$$\int \frac{x^{2009}}{(1+x^2)^{1006}} dx = \int \frac{x^{2009}}{\left(\frac{1}{x^2} + 1\right)^{1006} \cdot x^{2012}} dx$$

$$= \int \frac{\frac{1}{x^3} \cdot dx}{\left(\frac{1}{x^2} + 1\right)^{1006}} = -\frac{1}{2} \int \frac{\frac{-2}{x^3} dx}{\left(\frac{1}{x^2} + 1\right)^{1006}} = -\frac{1}{2} \frac{\left(\frac{1}{x^2} + 1\right)^{-1005}}{(-1005)} + C$$

$$= \frac{1}{2010} \left(\frac{x^2}{1+x^2} \right)^{1005} + C \Rightarrow \text{Comparing}$$

$$n=2010, m=1005 \Rightarrow m+n = \boxed{3015}$$

Ques Integrate $\int \frac{1+x \cos(x)}{x(1-x^2 e^{2 \sin(x)})} \cdot dx$

Sol

$$\int \frac{1+x \cdot \cos(x)}{x \cdot (1-x^2 \cdot e^{2 \sin(x)})} \cdot dx = \int \frac{(1+x \cos(x)) \cdot e^{\sin(x)}}{x \cdot e^{\sin(x)} (1-(x e^{\sin(x)})^2)} \cdot dx$$

$$= \int \frac{dt}{t(1-t^2)} = \int \frac{dt}{t^3 \left(\frac{1}{t^2} - 1 \right)}$$

$x \cdot e^{\sin(x)} = t$
 $(e^{\sin(x)} + x \cdot \cos(x) \cdot e^{\sin(x)}) dx = dt$

$$= \frac{1}{-2} \int \frac{\left(\frac{-2}{t^3} \right) dt}{\frac{1}{t^2} - 1} = -\frac{1}{2} \ln \left| \frac{1}{t^2} - 1 \right| + C = -\frac{1}{2} \ln \left| \frac{1}{x^2 e^{2 \sin(x)}} - 1 \right| + C.$$

Ans

Ques Integrate $\int \frac{x^4-1}{x^2 \sqrt{x^4+x^2+1}} \cdot dx$

Sol

$$\begin{aligned} \int \frac{x^4-1}{x^2 \sqrt{x^4+x^2+1}} \cdot dx &= \int \frac{(x^2-1)(x^2+1)}{x^2 \sqrt{x^4+x^2+1}} \cdot dx \\ &= \int \frac{\left(1-\frac{1}{x^2}\right) \cdot (x^2+1)}{x \cdot \sqrt{x^2+1+\frac{1}{x^2}}} \cdot dx = \int \frac{\left(x+\frac{1}{x}\right) \left(1-\frac{1}{x^2}\right)}{\sqrt{\left(x+\frac{1}{x}\right)^2 - 1}} \cdot dx \\ &= \sqrt{\left(x+\frac{1}{x}\right)^2 - 1} + C \quad \underline{A} \end{aligned}$$

Rough

$$(\sqrt{x})' = \frac{1}{2\sqrt{x}}$$

$$\left(\sqrt{x^2 \pm 1}\right)' = \frac{x}{\sqrt{x^2 \pm 1}}$$

Ques Find $\int (x^6 + x^4 + x^2) \cdot (2x^4 + 3x^2 + 6)^{1/2} \cdot dx$

Sol

$$\begin{aligned} & \int (x^6 + x^4 + x^2) \cdot (2x^4 + 3x^2 + 6)^{1/2} \cdot dx \\ &= \int (x^5 + x^3 + x) \cdot (2x^6 + 3x^4 + 6x^2)^{1/2} \cdot dx \\ &= \frac{1}{12} \int \underbrace{(12x^5 + 12x^3 + 12x)}_{f'} \cdot \underbrace{(2x^6 + 3x^4 + 6x^2)^{1/2}}_f \cdot dx \\ &= \frac{1}{12} \frac{(2x^6 + 3x^4 + 6x^2)^{3/2}}{\left(\frac{3}{2}\right)} + C \quad \text{Ans} \end{aligned}$$

Que
Evaluate

$$\int \frac{(x - \sin(x))^{3/2}}{\sqrt{x}} \cdot \left(\frac{6x^2 \cdot \sin^2\left(\frac{x}{2}\right)}{x - \sin(x)} + 3x \right) dx$$

Soln

$$\int \frac{(x - \sin(x))^{3/2}}{\sqrt{x}} \left(\frac{3x^2(1 - \cos(x)) + 3x(x - \sin(x))}{x - \sin(x)} \right) dx$$

$$= \int \frac{\sqrt{x - \sin(x)}}{\sqrt{x}} \left(6x^2 - 3x(x \cos(x) + \sin(x)) \right) dx$$

$$= \int \frac{\sqrt{x^2 - x \sin(x)}}{x} \left(6x^2 - 3x(x \cos(x) + \sin(x)) \right) dx = \int \frac{\sqrt{\underbrace{x^2 - x \sin(x)}_f}}{f} \left(\underbrace{2x - (x \cos(x) + \sin(x))}_{f'} \right) dx$$

$$= \beta \cdot \frac{(x^2 - x \sin(x))^{3/2}}{\left(\frac{3}{2}\right)} + C$$

$$= 2(x^2 - x \cdot \sin(x))^{3/2} + C$$

A

Ques If $y(x-y)^2 = x$, and $\int \frac{dx}{x-3y} = \lambda \cdot \ln((x-y)^2 - 1) + C$ then find value of ' λ '.

Soln. $y(x-y)^2 = x \xrightarrow{\frac{d}{dx}} y'(x-y)^2 + 2y(x-y)(1-y') = 1$
 $y'((x-y)^2 - 2y(x-y)) = 1 - 2y(x-y)$

$\int \frac{dx}{x-3y} = \lambda \cdot \ln((x-y)^2 - 1)$
 $\xrightarrow{\frac{d}{dx}} \frac{1}{x-3y} = \lambda \left(\frac{2(x-y) \cdot (1-y')}{(x-y)^2 - 1} \right)$ (1)

$y' = \frac{1 - 2y(x-y)}{(x-y)(x-y-2y)} = \frac{1 - 2xy + 2y^2}{(x-y)(x-3y)}$
 $\rightarrow 1 - y' = 1 - \frac{1 - 2xy + 2y^2}{(x-y)(x-3y)} = \frac{x^2 - 4xy + 3y^2 - 1 + 2xy - 2y^2}{(x-y)(x-3y)}$
 $\cdot 1 - y' = \frac{x^2 - 2xy + y^2 - 1}{(x-y)(x-3y)} = \frac{(x-y)^2 - 1}{(x-y)(x-3y)}$

$$1-y' = \frac{(x-y)^2 - 1}{(x-y)(x-3y)}$$

$$\frac{(x-y)(1-y')}{(x-y)^2 - 1} = \frac{1}{x-3y}$$

from ① $\frac{1}{x-3y} = \frac{\lambda \cdot 2 \cdot (x-y)(1-y')}{(x-y)^2 - 1} \Rightarrow \text{comparing}$

$$2\lambda = 1$$

②

$$\boxed{\lambda = \frac{1}{2}} \underline{\text{Ans.}}$$

Que

Evaluate $\int \frac{x^3 + x^2 + x - 1}{(x^2 - 1) \sqrt{2 - x^2}} \cdot dx$

Sol

$$\int \frac{x^3 + x^2 + x - 1}{(x^2 - 1) \sqrt{2 - x^2}} \cdot dx = \int \frac{x(x^2 - 1) + (x^2 - 1) + 2x}{(x^2 - 1) \sqrt{2 - x^2}} dx$$

$$= \int \frac{x}{\sqrt{2 - x^2}} \cdot dx + \int \frac{dx}{\sqrt{2 - x^2}} + \int \frac{dx}{(x+1)\sqrt{2 - x^2}} + \int \frac{dx}{(x-1)\sqrt{2 - x^2}}$$

$$= - \int \frac{\frac{f'}{f}}{2\sqrt{2 - x^2}} \cdot dx + \sin^{-1}\left(\frac{x}{\sqrt{2}}\right)$$

$$= - \sqrt{2 - x^2} + \sin^{-1}\left(\frac{x}{\sqrt{2}}\right) + f(x) + g(x) + c$$

$$\downarrow x+1 = \frac{1}{t} \Rightarrow u = \frac{1}{t} - 1 = \frac{1-t}{t}$$
$$dx = -\frac{dt}{t^2}$$

$$= \int \frac{-dt}{\frac{1}{t^2} \sqrt{\frac{t^2 + 2t - 1}{t^2}}}$$

$$2 - x^2 = 2 - \left(\frac{1-t}{t}\right)^2$$
$$= \frac{t^2 + 2t - 1}{t^2}$$

$$\begin{aligned} f(x) &= - \int \frac{dt}{\sqrt{t^2 + 2t - 1}} = - \int \frac{dt}{\sqrt{(t+1)^2 - 2}} \\ &= - \ln |t+1 + \sqrt{(t+1)^2 - 2}| + C \end{aligned}$$

Ques Integrate $\int \frac{\sqrt{\cot(x)} - \sqrt{\tan(x)}}{1 + 3\sin(2x)} \cdot dx$

Soln $\int \frac{\sqrt{\cot(x)} - \sqrt{\tan(x)}}{1 + 3\sin(2x)} \cdot dx = \int \frac{\cos(x) - \sin(x)}{\sqrt{\sin(x)\cos(x)} \cdot (1 + 3\sin(2x))} \cdot dx$

$$= \sqrt{2} \int \frac{c - s}{\sqrt{\sin(2x)} (1 + 3\sin(2x))} \cdot dx = \sqrt{2} \int \frac{c - s}{\sqrt{(s+c)^2 - 1} \cdot (1 + 3(s+c)^2 - 1))} \cdot dx$$

$$= \sqrt{2} \int \frac{c - s}{\sqrt{(s+c)^2 - 1} (3(s+c)^2 - 2)} \cdot dx$$

$s + c = t \Rightarrow (c - s) dx = dt$

$$= \sqrt{2} \int \frac{dt}{\sqrt{t^2 - 1} (3t^2 - 2)}$$

$$= \sqrt{z} \int \frac{dA}{(3t^2-2)\sqrt{t^2-1}}$$

$$= \sqrt{z} \int \frac{\left(\frac{dA}{t^3}\right)}{\left(3-\frac{2}{t^2}\right)\sqrt{1-\frac{1}{t^2}}}$$

$$= \sqrt{z} \int \frac{\cancel{z} \cdot dz}{(1+2z^2) \cdot \cancel{z}} = \frac{1}{\sqrt{z}} \int \frac{dz}{z^2 + \frac{1}{z}}$$

$$= \frac{1}{\cancel{\sqrt{z}}} \cdot \frac{1}{\left(\frac{1}{\sqrt{z}}\right)} \tan^{-1}\left(\frac{z}{\left(\frac{1}{\sqrt{z}}\right)}\right) + C = \tan^{-1}(\sqrt{z} z) + C$$

$$= \tan^{-1}\left(\sqrt{z} \sqrt{\frac{t^2-1}{t^2}}\right) + C = \tan^{-1}\left(\frac{\sqrt{z} \sqrt{(5+1)^2-1}}{5+1}\right) + C$$

$$1 - \frac{1}{t^2} = z^2 + \frac{2}{t^3} \cdot dt = d\left(z + \frac{1}{t^2}\right)$$

$$3 - \frac{2}{t^2} = 1 + 2\left(1 - \frac{1}{t^2}\right) = 1 + 2z^2$$

$$\tan^{-1}\left(\frac{\sqrt{z} \sin(2x)}{\sin(x) + \cos(x)}\right)$$

+ C

Que Evaluate $\int \frac{x + e^x (\sin x + \cos x) + \sin x \cdot \cos x}{(x^2 + 2e^x \sin x - \cos^2 x)^2} \cdot dx$

Sol

$$\int \frac{x + e^x (\sin x + \cos x) + \sin x \cdot \cos x}{(x^2 + 2e^x \sin x - \cos^2 x)^2} \cdot dx$$

$$= \frac{1}{2} \int \frac{2x + 2e^x (\sin x + \cos x) + 2 \sin x \cdot \cos x}{\underbrace{(x^2 + 2e^x \sin x - \cos^2 x)}_f^2} \cdot dx$$

$$= -\frac{1}{2} \cdot \frac{1}{x^2 + 2e^x \sin x - \cos^2 x} + C$$

$$(e^x \cdot f)' = e^x (f + f')$$

Que Evaluate

$$\int x \cdot \sqrt{\frac{2 \sin(x^2+1) - \sin(2(x^2+1))}{2 \sin(x^2+1) + \sin(2(x^2+1))}} \cdot dx$$

$$\text{Let } x^2+1 = \theta \\ 2x dx = d\theta$$

Sol

$$\int x \cdot \sqrt{\frac{2 \sin(x^2+1) - \sin(2(x^2+1))}{2 \sin(x^2+1) + \sin(2(x^2+1))}} \cdot dx$$
$$= \frac{1}{2} \int \frac{2 \sin(\theta) - \sin(2\theta)}{2 \sin(\theta) + \sin(2\theta)} d\theta = \frac{1}{2} \int \sqrt{\frac{1 - \cos(\theta)}{1 + \cos(\theta)}} d\theta = \frac{1}{2} \int \sqrt{\tan^2\left(\frac{\theta}{2}\right)} d\theta = \frac{1}{2} \int \tan\left(\frac{\theta}{2}\right) d\theta$$
$$= \frac{1}{2} \frac{\ln|\sec(\frac{\theta}{2})|}{(\frac{1}{2})} + C = \ln\left|\sec\left(\frac{x^2+1}{2}\right)\right| + C$$

An

Que Evaluate $\int e^x \left(\frac{x^4 + 2}{(1+x^2)^{5/2}} \right) \cdot dx$

Sol

$$\int e^x \left(\frac{x^4 + 2}{(1+x^2)^{5/2}} \right) \cdot dx$$

$$= \int e^x \left(\frac{x^4 + 2x^2 + 1 + 1 - 2x^2}{(1+x^2)^{5/2}} \right) \cdot dx$$

$$= \int e^x \left(\frac{(1+x^2)^2 + 1 - 2x^2}{(1+x^2)^{5/2}} \right) \cdot dx$$

C-2

$$\left(\frac{1}{f^3} \right)' = -\frac{3 \cdot f'}{f^4}$$

C-3

$$\frac{(x+2)^7}{(x+2)^7} \sim \frac{(x+2)^7}{(x+2)^7}$$

C-3

$$\int e^x (f - f'') \cdot dx$$

C-1

$$\int e^x (f + f') \cdot dx$$

$$= \int (e^x \cdot f + e^x \cdot f') \cdot dx$$

$$= \int (e^x \cdot f)' \cdot dx$$

$$= e^x \cdot f(x)$$

C-1

$$\int e^x \cdot f \cdot dx + \int e^x \cdot f' \cdot dx$$

C-1

$$e^x \cdot f - \int e^x \cdot f' \cdot dx$$

$$= \int e^x \left(\frac{1}{\sqrt{x^2+1}} + \frac{1-2x^2}{(x^2+1)^{5/2}} \right) \cdot dx$$

$$= \int e^x \left(\frac{1}{\sqrt{x^2+1}} - \frac{x}{(x^2+1)^{3/2}} + \frac{x}{(x^2+1)^{3/2}} + \frac{1-2x^2}{(x^2+1)^{5/2}} \right) \cdot dx$$

$$= e^x \cdot \frac{1}{\sqrt{x^2+1}} + e^x \cdot \frac{x}{(x^2+1)^{3/2}}$$

+ C

Ans

$$\left((x^2+1)^{-1/2} \right)' = -\frac{1}{2} (x^2+1)^{-3/2} \cdot 2x$$

$$= -\frac{x}{(x^2+1)^{3/2}}$$

$$\left(\frac{x}{(x^2+1)^{3/2}} \right)' = \frac{(x^2+1)^{3/2} - x \cdot \frac{3}{2} \cdot (x^2+1)^{1/2} \cdot 2x}{(x^2+1)^3}$$

$$= \frac{x^2+1-3x^2}{(x^2+1)^{5/2}} = \frac{1-2x^2}{(x^2+1)^{5/2}}$$

Rough

Ques Evaluate $\int e^{x \sin(x) + \cos(x)} \left(\frac{x^2 \cos^2(x) - x \sin(x) - \cos(x)}{x^2} \right) \cdot dx$

Sol

$$\int e^{x \sin(x) + \cos(x)} \left(\frac{x^2 \cos^2(x) - x \sin(x) - \cos(x)}{x^2} \right) \cdot dx$$

$$= \int e^{\overbrace{x \sin(x) + \cos(x)}^g} \left(\cos^2(x) - \left(\frac{x \sin(x) + \cos(x)}{x^2} \right) \right) \cdot dx$$

$$g' = (x \sin(x) + \cos(x))' = x \cos(x) + 1 - \cancel{x} - \cancel{1}$$

$$= x \cos(x)$$

Rough

$$\int e^g (g' \cdot f + f') \cdot dx$$

$$= \int (e^g \cdot f)' dx = e^g \cdot f + C$$

$$e^x (f + f')$$

$$\int e^g \left(\boxed{x \cos(x)} + \cancel{\frac{1}{x}} \right) dx$$

$$= \int e^{\overbrace{x \sin(x) + \cos(x)}^g} \left(\cos^2(x) - \left(\frac{x \sin(x) + \cos(x)}{x^2} \right) \right) dx$$

$$= \int e^{\overbrace{x \sin(x) + \cos(x)}^g} \left(\underbrace{(x \cdot \cos(x))}_{g'} \underbrace{\left(\frac{\cos(x)}{x} \right)}_f + \underbrace{\left(\frac{-x \sin(x) - \cos(x)}{x^2} \right)}_{f'} \right) dx$$

$$= e^{x \sin(x) + \cos(x)} \cdot \frac{\cos(x)}{x} + C$$

$$= \frac{\cos(x) \cdot e^{x \sin(x) + \cos(x)}}{x} + C \quad \underline{\text{Ans}}$$

Proof

$$\int e^x \left(\frac{\sin(x)}{f'} - \frac{\cos(x)}{f} \right) dx$$
$$= -\cos(x) \cdot e^x + C$$